Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Eksamen

Examination

Module EBN122

Elektrisiteit en Elektronika

15 November 2010

Module EBN122

Electricity and Electronics

15 November 2010

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksameninligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	180 minute *180 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		Vraestel/ *Exam paper*: 21 Antwoordblad/ *Answer sheet*: 2	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All Questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a+jb$
   Polar form: $A\angle\theta$ *; Rectangular form:* $a+jb$

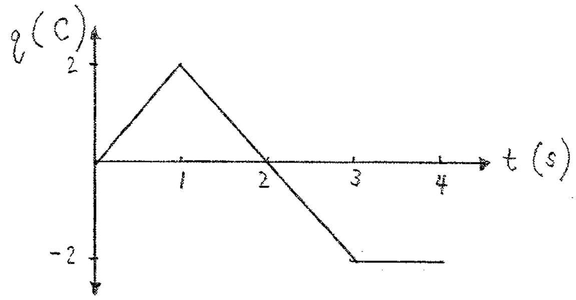
Dosent(e): *Lecturer(s):*	1 JP Jacobs 2 B Mabuza 3 JD le Roux

Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groephoof: *Group Head:*	Prof JW Odendaal	Handtekening: *Signature:*	

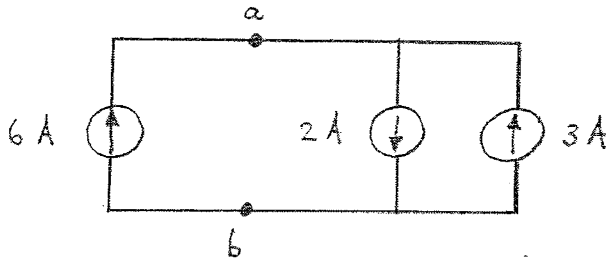
VRAAG/QUESTION 1 [30]

Vraag 1.1 [2]	Question 1.1 [2]
Die lading deur ’n draad word hieronder getoon as ‘n funksie van tyd. Skets die stroom gedurende die tydinterval $0 \leq t \leq 4$ s.	The charge through a wire is shown below as a function of time. Sketch the current for the time period $0 \leq t \leq 4$ s.

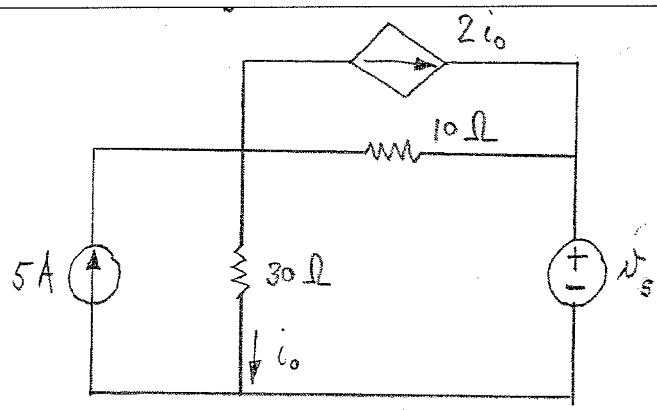


Vraag 1.2 [2]	Question 1.2 [2]
Die stroom wat die positiewe terminal van ‘n toestel binnegaan, is $i(t) = 4e^{-4t}$ A en die spanning oor die toestel is $v(t) = 2di/dt$ V. Bepaal die energie geabsorbeer tussen $t = 0$ s en $t = 2$ s.	The current entering the positive terminal of a device is $i(t) = 4e^{-4t}$ A and the voltage over the device is $v(t) = 2di/dt$ V. Find the energy absorbed between $t = 0$ s and $t = 2$ s.

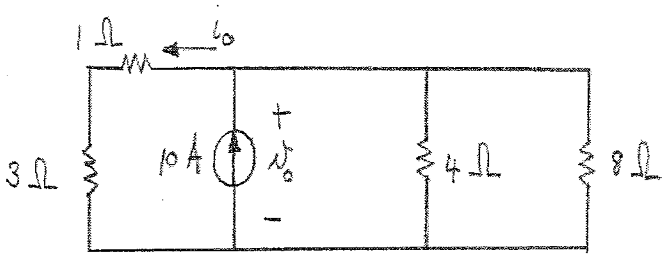
Vraag 1.3 [1]	Question 1.3 [1]
Skets 'n ekwivalente stroombaan met betrekking tot terminale *a-b* wat slegs een stroombron sal bevat.	Sketch an equivalent circuit with respect to terminals *a-b* that will contain only one current source.



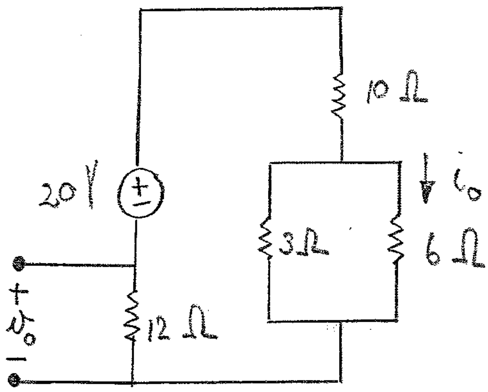
Vraag 1.4 [5]	Question 1.4 [5]
Vir die stroombaan hieronder is dit bekend dat $i_o = 2$ A. Bepaal: a) die drywing geassosieer met die onafhanklike spanningsbron [2] b) die drywing geassosieer met die onafhanklike stroombron [1] c) die drywing geassosieer met die afhanklike stroombron [1] d) die totale drywing gedissipeer in die twee weerstande [1]	For the circuit below it is known that $i_o = 2$ A. Determine: a) the power associated with the independent voltage source [2] b) the power associated with the independent current source [1] c) the power associated with the dependent current source [1] d) the total power dissipated in the two resistors [1]



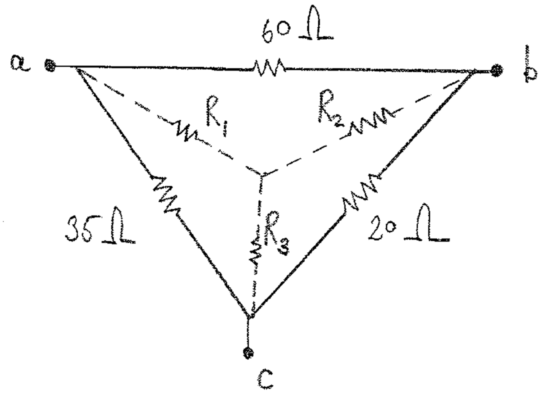
Vraag 1.5 [2]	Question 1.5 [2]
Bepaal v_o en i_o.	Determine v_o and i_o.



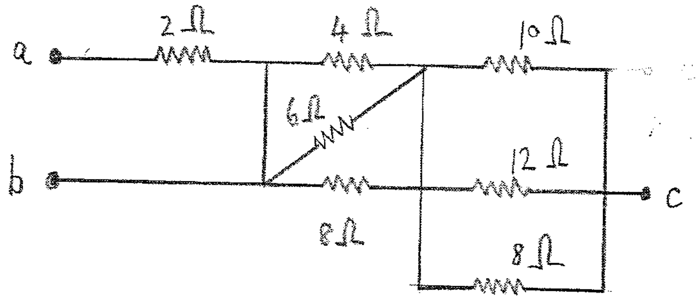
Question 1.6 [2]	Question 1.6 [2]
Bepaal v_o en i_o deur van spanningsverdeling gebruik te maak.	Determine v_o and i_o by using voltage division.



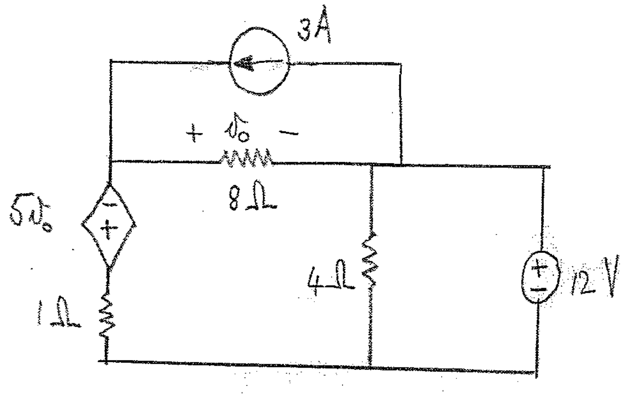
Vraag 1.7 [3]	Question 1.7 [3]
Skakel die Δ–netwerk om na ‘n ekwivalente Y-netwerk. Bepaal R_1, R_2 en R_3.	Convert the Δ network to an equivalent Y network. Determine R_1, R_2 and R_3.



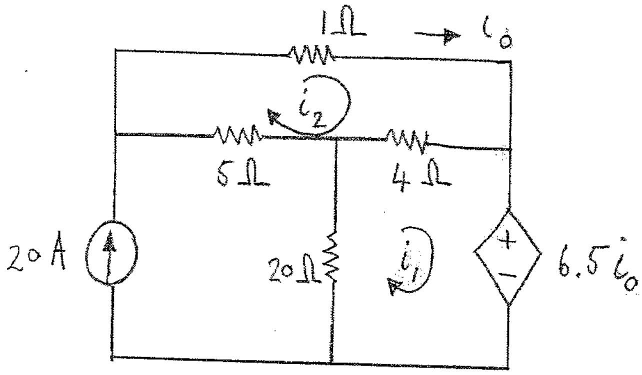
Vraag 1.8 [4]	Question 1.8 [2]
Bepaal die ekwivalente weerstand gesien deur terminale *b*-*c*.	Find the equivalent resistance seen through terminals *b*-*c*.



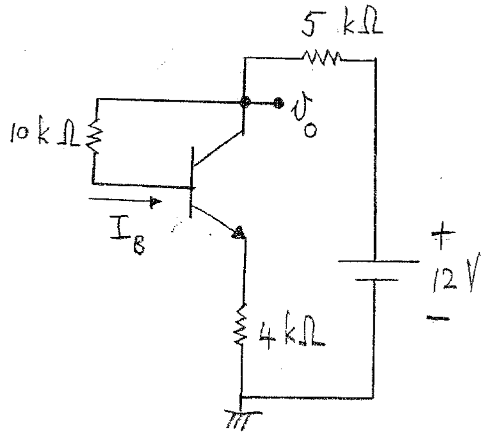
Vraag 1.9 [3]	Question 1.9 [3]
Gebruik node-analise om v_o te bepaal.	Use nodal analysis to determine v_o.



Vraag 1.10 [4] Gebruik die lusstroom-metode om twee vergelykings van die volgende vorm te formuleer: $ai_1 + bi_2 = 40$ $ci_1 + di_2 = 100$	**Question 1.10 [4]** Use the mesh current method to formulate two equations of the form $ai_1 + bi_2 = 40$ $ci_1 + di_2 = 100$

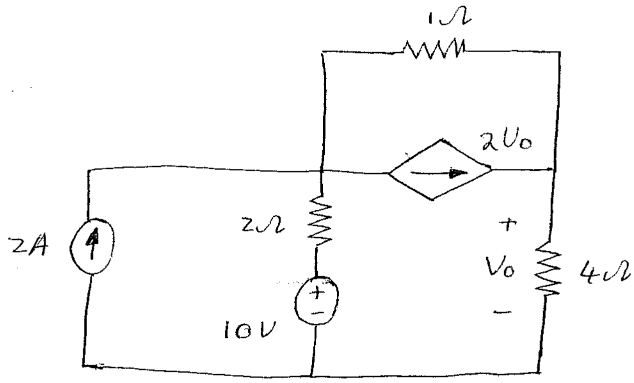


Vraag 1.11 [2] Bepaal I_B (in μA) indien $\beta = 100$ en $V_{BE} = 0.7$ V. *Wenk*: gebruik KVL.	**Question 1.11 [2]** Determine I_B (in μA) when $\beta = 100$ and $V_{BE} = 0.7$ V. *Hint*: use KVL.

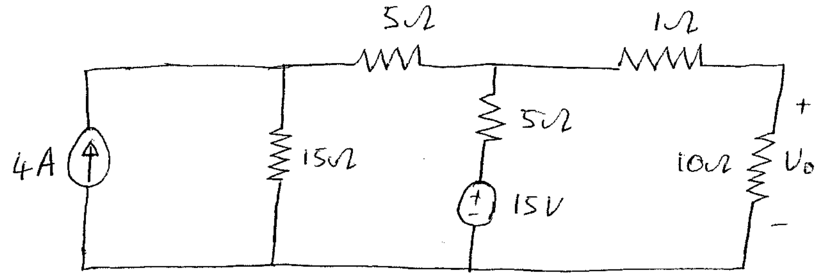


VRAAG/QUESTION 2 [30]

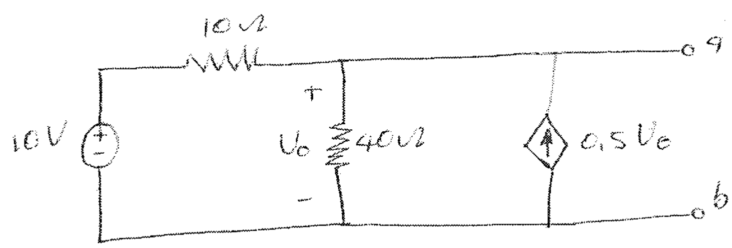
Vraag 2.1 [4]	Question 2.1 [4]
a) Bepaal V_0 as gevolg van die 10 V spanningsbron. b) Bepaal V_0 as gevolg van die 2 A stroombron.	a) Find V_0 due to the 10 V voltage source. b) Find V_0 due to the 2 A current source.



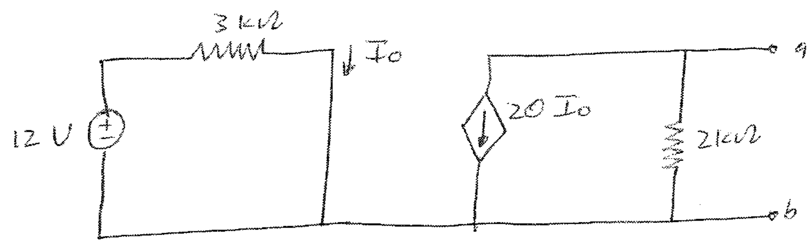
Vraag 2.2 [2] Gebruik brontransformasie om V_0 te bereken.	**Question 2.2 [2]** Use source transformation to find V_0.



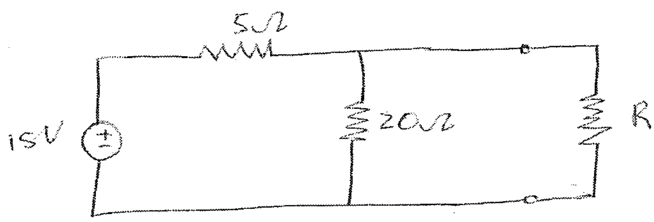
Vraag 2.3 [4]	**Question 2.3 [4]**
Bepaal die Thevenin ekwivalent met betrekking tot terminale a-b.	Find the Thevenin equivalent at terminal a-b.



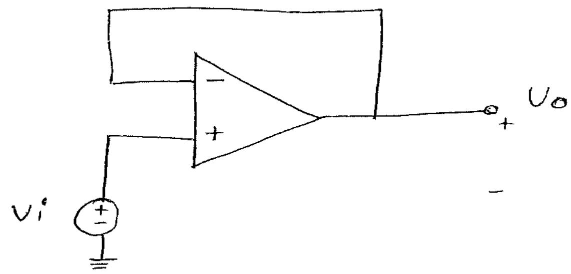
Vraag 2.4 [4] Bepaal die Norton ekwivalent met betrekking tot terminale a-b.	**Question 2.4 [4]** Find the Norton equivalent at terminals a-b.



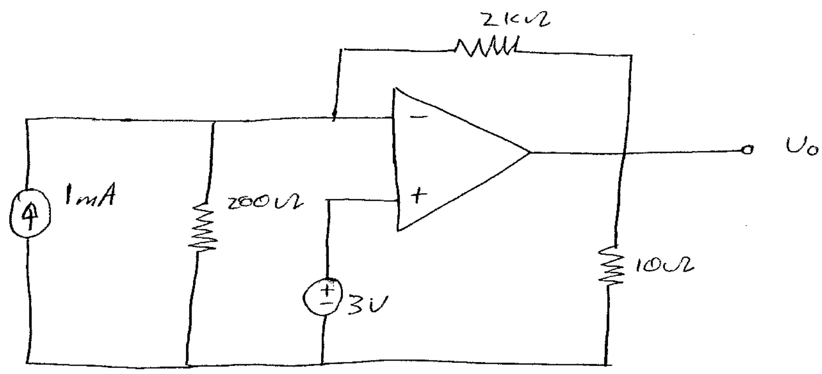
Vraag 2.5 [4] Bereken: a) *R* vir maksimum drywingsoordrag. b) Die maksimum drywing gelewer aan *R*.	**Question 2.5 [4]** Calculate: a) *R* for maximum power transfer. b) The maximum power delivered to *R*.



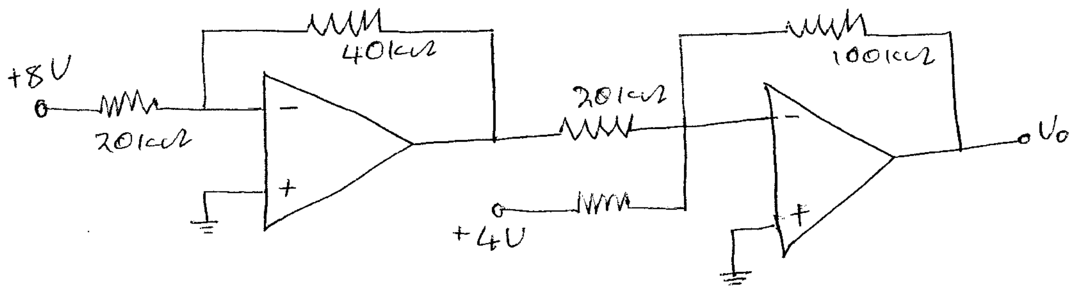
Vraag 2.6 [4]	Question 2.6 [4]
Bepaal: a) V_0/V_i indien $R_i = 100\ \text{k}\Omega$, $R_0 = 100\ \Omega$ en $A = 100{,}000$ **(nie-ideale OV)**. *b)* V_0/V_i indien $R_i = \infty\ \Omega$, $R_0 = 0\ \Omega$ en $A = \infty$ **(ideale OV)**.	Find: a) V_0/V_i if $R_i = 100\ \text{k}\Omega$, $R_0 = 100\ \Omega$ and $A = 100{,}000$ **(non-ideal op-amp)**. b) V_0/V_i if $R_i = \infty\ \Omega$, $R_0 = 0\ \Omega$ and $A = \infty$ **(ideal op-amp)**.



Vraag 2.7 [4] Bepaal: a) V_0 as gevolg van die 1 mA stroombron (deaktiveer die spanningsbron). b) V_0 as gevolg can die 3 V spanningsbron (deaktiveer die stroombron).	**Question 2.7 [4]** Find: a) V_0 due to the 1 mA current source (turn off the voltage source). b) V_0 due to the 3 V voltage source (turn off the current source).

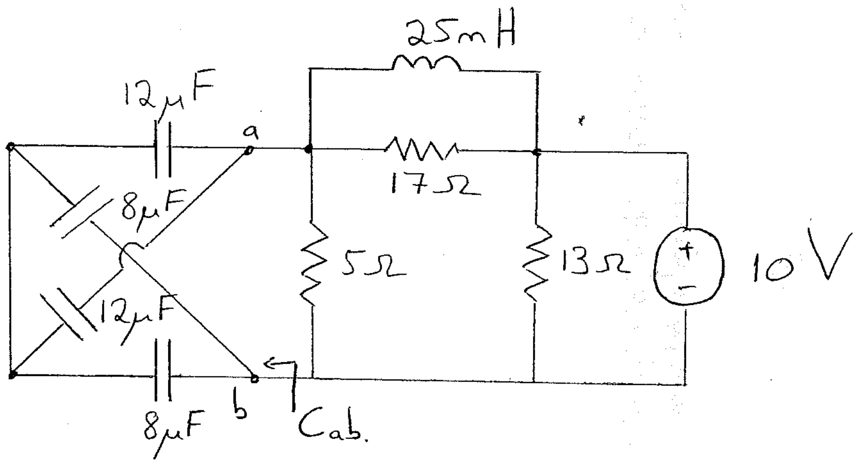


Vraag 2.8 [4]	Question 2.8 [4]
a) Watter tipe versterkerbaan is die eerste fase? b) Bepaal V_0.	a) What type of amplifier circuit is the first stage? b) Find V_0.



VRAAG/QUESTION 3 [30]

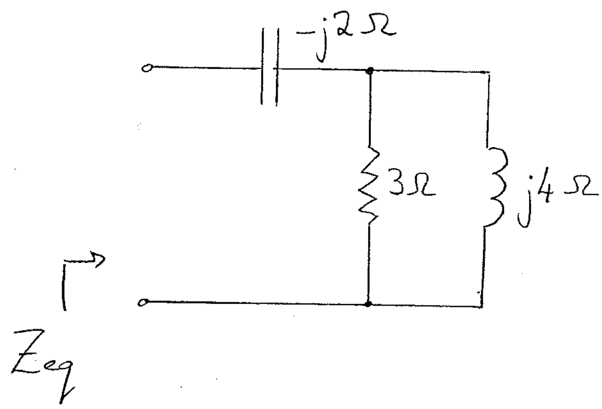
Vraag	Question
Vraag 3.1 [2] Die spanning oor 'n 10 mH induktor is $8e^{-2t}$ mV vir $t \geq 0$ s. Bepaal die stroom *i(t)* deur die induktor. Aanvaar dat $i(0) = 1.6$ A	**Question 3.1 [2]** The voltage across a 10 mH inductor is $8e^{-2t}$ mV for $t \geq 0$ s. Determine the current *i(t)* through the inductor. Assume that $i(0) = 1.6$ A
Vraag 3.2 [3] a) Bepaal die ekwivalente kapasitansie (C_{ab}) oor terminale a-b. [1] b) Bepaal die energie gestoor in die induktor vir gelykstroom toestande. [2]	**Question 3.2 [3]** a) Determine the equivalent capacitance (C_{ab}) across terminals a-b. [1] b) Determine the energy stored in the inductor for DC conditions. [2]



Vraag 3.3 [2]	Question 3.3 [2]
Skryf as 'n enkele cosinus uitdrukking: $v(t) = 7.07\cos(\omega t - 45°) + 10\sin(\omega t - 90°)$ V	Write as a single cosine expression: $v(t) = 7.07\cos(\omega t - 45°) + 10\sin(\omega t - 90°)$ V

Vraag 3.4 [2]	Question 3.4 [2]
Watter van die volgende seine loop voor en teen watter hoek? $\overline{V_1} = 10\angle -30°$, $v_2(t) = -2\sin(\omega t - 45°)$	Which of the following signals is leading and by what angle? $\overline{V_1} = 10\angle -30°$, $v_2(t) = -2\sin(\omega t - 45°)$

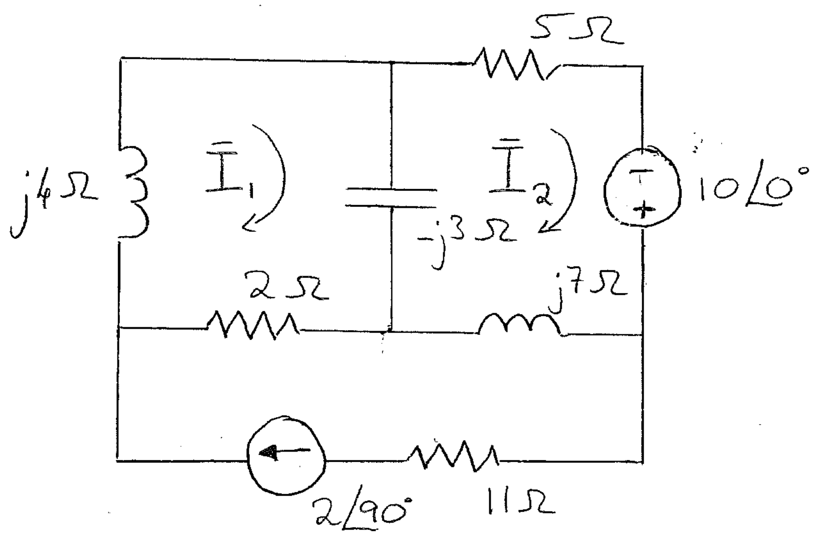
Vraag 3.5 [2]	Question 3.5 [2]
Bepaal Z_{eq} (in reghoekige vorm):	Determine Z_{eq} (in rectangular form):



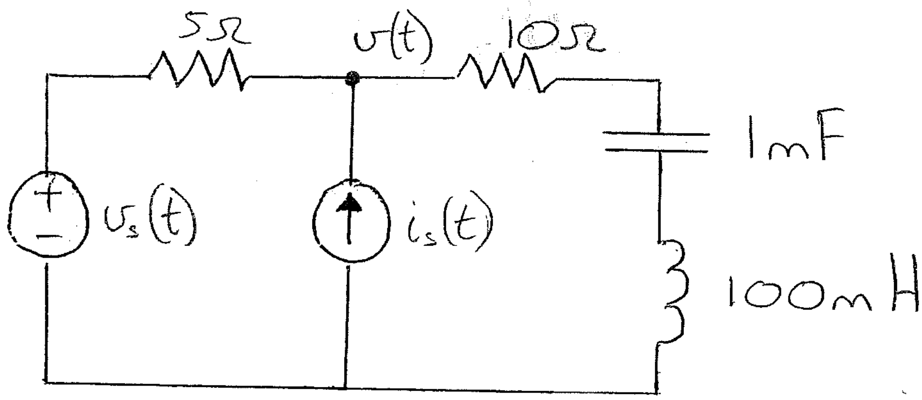
Vraag 3.6 [2]	**Question 3.6 [2]**
Evalueer die volgende (skryf antwoord in polêre vorm): $$\left(\frac{1}{4\sqrt{2}}-j\frac{1}{4\sqrt{2}}\right)^{-\frac{1}{2}}$$	Evaluate the following (write the answer in polar form): $$\left(\frac{1}{4\sqrt{2}}-j\frac{1}{4\sqrt{2}}\right)^{-\frac{1}{2}}$$

Vraag 3.7 [2]	**Question 3.7 [2]**
Bepaal $\overline{V}_1$ (in reghoekige vorm) indien $\overline{V}_2 = 6$ V: $$\frac{\overline{V}_1-2}{-5+j9}+\frac{\overline{V}_2}{-j3}=0$$	Determine $\overline{V}_1$ (in rectangular form) if $\overline{V}_2 = 6$ V: $$\frac{\overline{V}_1-2}{-5+j9}+\frac{\overline{V}_2}{-j3}=0$$

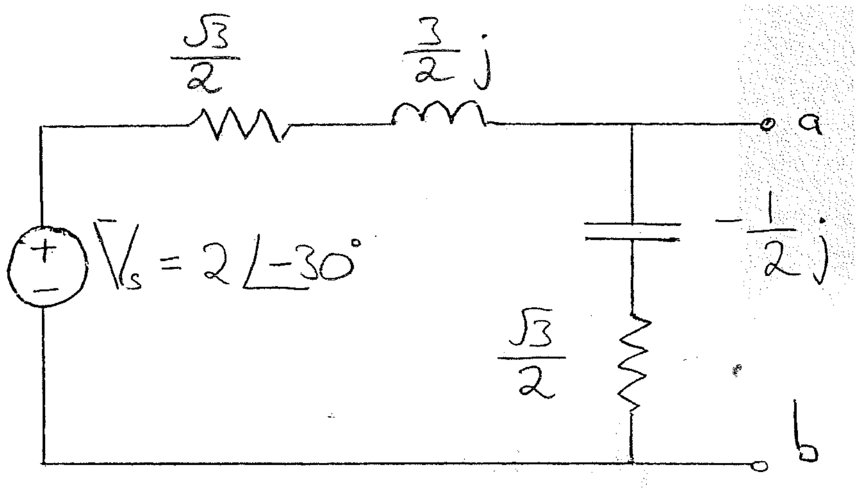
Vraag 3.8 [4]	**Question 3.8 [4]**
Gebruik lus-analise om twee gelyktydige vergelykings vir fasors $\bar{I}_1$ en $\bar{I}_2$ op te stel. Skryf in die vorm $\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = 4j$ $\bar{c}\bar{I}_1 + \bar{d}\bar{I}_2 = -4$ met $\bar{a}$, $\bar{b}$, $\bar{c}$, en $\bar{d}$ komplekse koeffisiënte in reghoekige vorm.	Use mesh analysis to set up two simultaneous equations for phasors $\bar{I}_1$ and $\bar{I}_2$. Write in the form $\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = 4j$ $\bar{c}\bar{I}_1 + \bar{d}\bar{I}_2 = -4$ with $\bar{a}$, $\bar{b}$, $\bar{c}$, and $\bar{d}$ complex coefficients in rectangular form.



Vraag 3.9 [4]	**Question 3.9 [4]**
Gebruik die beginsel van superposisie. a) Bereken node spanning *v(t)* in **fasorvorm** ($\overline{V}$) slegs as gevolg van die spanningsbron $v_s(t) = 10\sqrt{2}\cos(200t + 45^{\circ})$ V b) Bereken *v(t)* as 'n **tydsein** slegs as gevolg van die stroombron $i_s(t) = 5\cos(100t)$ A	Use the principle of superposition. a) Determine node voltage *v(t)* in **fasor form** ($\overline{V}$) only due to the voltage source $v_s(t) = 10\sqrt{2}\cos(200t + 45^{\circ})$ V b) Determine *v(t)* as a **time signal** only due to the current source $i_s(t) = 5\cos(100t)$ A



Vraag 3.10 [4]	**Question 3.10 [4]**
Bepaal die Thevenin ekwivalent ten opsigte van terminale a-b.	Find the Thevenin equivalent with respect to terminals a-b.



Vraag 3.11 [3]	**Question 3.11 [3]**
a) Bepaal die verhouding $\overline{V_O}/\overline{V_S}$ in terme van *C, R* en ω.	a) Determine the relationship $\overline{V_O}/\overline{V_S}$ in terms of *C, R* and ω.
b) Indien $v_S(t) = 10\cos(10t)$ V, $R = 1\ \Omega$ en $C = 0.1$ F, bepaal $v_O(t)$.	b) If $v_S(t) = 10\cos(10t)$ V, $R = 1\ \Omega$ and $C = 0.1$ F, determine $v_O(t)$.